# Tower Bridge - Kids

UK Family Tour

## A Bridge That Opens Like a Drawbridge

Tower Bridge is the most famous bridge in the world that people call by the wrong name — lots of visitors think it's London Bridge. It's not! London Bridge is the plain one just upriver. This one, with the two fairy-tale towers, opened in 1894, and it has a superpower: the road splits in the middle and lifts up like two giant see-saws to let tall ships through. Each half of the road weighs over a thousand tonnes — as much as two hundred elephants — yet it takes only about a minute to rise. The bridge still opens around eight hundred times a year, and here's the best rule ever: boats beat cars. River traffic has priority, always, and lifting the bridge is completely free — a ship just has to book twenty-four hours ahead. Even the President of the United States once had to wait: in 2009, President Clinton's motorcade got split in half when the bridge opened for a barge, right on schedule. The bridge didn't apologize. Its colors — blue, white and red — aren't original, by the way. It was painted these colors in 1977 for the Queen's Silver Jubilee. Before that it was chocolate brown.

### A Castle Costume Over Iron Bones

Look up at the two great towers. They look like they belong in a fairy tale, don't they, with grey stone walls, pointy tops, and little windows where you could imagine a knight standing guard. Here is the secret nobody tells you. It is all a costume. Underneath that stone, the towers are built from steel, a skeleton of iron beams as strong as the frame of a skyscraper. The stone is not really holding the bridge up at all. It is more like a heavy coat, buttoned over the iron bones to make the whole thing beautiful.

So why dress a modern machine up like an old castle? Because of the neighbour. Right next door stands the Tower of London, a real fortress that has guarded the river for almost a thousand years, ever since William the Conqueror. When this bridge was being planned in the eighteen eighties, the people in charge worried that a plain iron bridge would look rude and ugly sitting beside something so ancient. So they made a rule. The new bridge had to match. It had to look like it belonged. Dress up smartly, or you don't get built.

And so the engineers gave their steel monster a disguise. They wrapped the towers in stone and granite, added arches and turrets and all the pointy bits on top, and made the whole thing look serious and old. The trick worked so well that today almost everyone is fooled. They think Tower Bridge is hundreds of years old, from the days of kings and knights. It is not. When it opened in eighteen ninety-four, it was one of the most modern, cleverest machines on the entire planet. Think of it as a giant robot bridge, dressed up as a castle for a costume party, and never taking the costume off.

So next time somebody tells you the bridge looks medieval, you can smile and tell them the truth. Those stone walls are not doing the heavy work. They are a splendid Victorian trick. The real muscle is hidden steel, and the real magic is waiting for you downstairs, in the engine rooms. The castle is only the outfit.

## Watch an Opening Happen

Imagine you are standing on the bridge and an opening is about to begin. It does not just fly up out of nowhere. There is a careful little ceremony first, and once you know the steps, you can watch for them.

First, the traffic lights turn red and the cars stop. Then barriers swing across the road, like the gates at a railway crossing, so nobody can drive on. A bell rings, and a voice may warn people to step back behind the line. Only when the bridge is certain that every car and every person is safely off the moving part does the real show begin. There is even a person whose whole job is to make this happen, watching from a control room high in the bridge, checking that the road is clear before pressing the button that sets a thousand tonnes of steel in motion.

Then, slowly, the two halves of the road begin to tip upward. They rise and rise until they point almost straight at the sky, opening a huge gap in the middle for a ship to glide through. When the boat has passed, the halves come back down, meet exactly in the middle, lock together, the barriers lift, the lights turn green, and the cars roll on as if nothing amazing just happened.

Here is something surprising about the machinery. For the bridge's first eighty years or so, all of that lifting was powered by steam, made by burning coal, just like an old railway engine. But in nineteen seventy-six the bridge got a modern makeover. Now the great arms are moved by electric motors pushing oil through pipes, which is cleaner, quieter, and does not need anyone shovelling coal all night. The bridge kept its castle costume on the outside, but swapped its old steam heart for a new one.

So if you are lucky enough to catch an opening while you are here, do not just watch the road lift. Watch the whole ceremony. The red lights, the barriers, the bell, the pause, and then the giant see-saws climbing toward the clouds. It is one of the best free shows in all of London, and it has been playing, on and off, for well over a hundred years.

## Walking on Glass, 42 Meters Up

Ready for the scary-fun part? Up on the high walkways between the towers, part of the floor is made of glass, and you can stand on it and look straight down — forty-two meters, about the height of a ten-story building — to the road and river below. Cars and red buses zoom underneath your shoes. Don't worry: each glass panel is made of five layers and can hold the weight of an elephant, or about seven London black cabs. The engineers tested it properly. If you time it right and the bridge opens while you're up there, you'll see the road split apart directly beneath your feet — one of the coolest views in London. Fun fact about these walkways: when the bridge was new, they were open so pedestrians could cross even when the road was lifted. But almost nobody used them — people preferred to wait and watch the ships — and the walkways became a hangout for pickpockets, so they were closed in 1910 for almost a hundred years. Now they're back, with glass floors and much better behavior. Lie down on the glass if you dare. Everyone does it. Take the photo.

### How They Made a Floor You Can Trust

Standing on glass forty-two meters up takes a bit of bravery, even when your brain knows it is safe. So how did the engineers make a floor that you really, truly can trust with your whole family standing on it at once? The answer is testing, and lots of it.

The glass floor was added in twenty fourteen, long after the walkways themselves were built. Each panel is not one sheet of glass, like a window. It is five separate layers, glued together into one thick, super-strong sandwich. If one layer ever cracked, the others would still hold, so the floor could never suddenly give way. And these panels are heavy. A single one weighs more than half a tonne, about the same as a grand piano. It takes several people just to lift one into place.

Before they let a single visitor step on, the engineers did what engineers love to do. They tried to break it, on purpose. They piled on huge weights, far heavier than any crowd of people could ever be, to prove the glass could take much more than it would ever need to. Only when it passed every test did they open it to you. So when you read that a panel could hold the weight of an elephant, that is not a wild guess. Somebody actually measured it.

Now here is a puzzle to think about. How do you clean a glass floor that hangs forty-two meters above a busy road and a river full of boats? You cannot exactly lean a ladder against the sky. The team who look after the bridge have special ways to reach every panel, wiping away all the scuffs from thousands of shoes and, yes, all the smudges from thousands of noses and hands pressed against the glass by people lying down to peer through. Because everyone does lie down. It is impossible to resist.

So go ahead. Trust the engineers, trust the five layers, trust the elephant test, and lie right down on the glass. Watch a red bus slide underneath your tummy. The people who built this floor spent months making absolutely certain that the only thing between you and the drop is glass strong enough to laugh at it.

### The Ghost Years of the High Walkways

The walkways you are standing on have a strange and slightly spooky history. For most of the twentieth century, they were shut, locked, empty, and forgotten. Imagine that. Two long bridges in the sky, way above one of the busiest cities on Earth, and hardly a soul allowed up here for over seventy years.

It started with a good idea that simply did not work. When the bridge opened in eighteen ninety-four, the walkways had a clever purpose. If you were walking across and the road lifted for a ship, you did not have to wait. You could climb the tower stairs, cross high up on the walkways, and come down the other side, never stopping. Sensible, right? Except that hardly anyone bothered. Climbing all those steps was tiring, and Victorians actually enjoyed waiting a few minutes to watch a great sailing ship pass by. Watching ships was free entertainment. Stairs were just stairs.

So the walkways sat almost empty, and empty places in a big city can attract trouble. High above the crowd, out of sight of any policeman, the lonely walkways became a favourite spot for pickpockets, thieves who quietly slip valuables out of people's pockets, and other people up to no good. The bridge decided enough was enough. In nineteen ten, they closed the walkways completely and locked the doors.

And then, for decade after decade, they stayed shut. Rain lashed the windows. Pigeons moved in. The whole city changed below, wars came and went, but the high walkways just waited in silence, like a secret nobody remembered. It was not until nineteen eighty-two, more than seventy years later, that they were cleaned up, made safe, and finally opened again, this time as part of the exhibition you are visiting today. The glass floor came even later, in twenty fourteen.

So think about that for a moment. The very spot where you are standing, taking your photo, was once a forbidden, empty ghost bridge that people whispered about. Now it is one of the most popular views in London, full of families and cameras and children lying flat on the glass. The pickpockets are long gone. The ghost years are over. And the walkways, at last, are doing the job they were built for, more than a hundred years late.

## The Day a Bus Jumped the Bridge

This really happened. On a December evening in 1952, a red double-decker bus, the number 78, was driving across Tower Bridge when the driver, Albert Gunter, felt the road tilting under his wheels. The bridge was opening — with his bus still on it! The warning gates hadn't closed properly. Albert had about one second to decide: brake and maybe slide backwards into the gap, or... floor it. He floored it. The bus roared up the rising road like a ramp and jumped the gap, landing with a crash on the other side, which hadn't started lifting yet. The bus flew about a meter down onto the far span. Everyone survived — twelve passengers, the conductor, and Albert, who broke... nothing. He got a ten-pound reward and a day off. When reporters asked him about it, he was very calm, and reportedly asked his bosses for something extra for his bravery. Down in the Victorian Engine Rooms, you'll see the giant coal-fired steam engines that powered the bridge for over eighty years — the same kind of machinery that was lifting the road under Albert's bus that night.

### The Man Who Flew Through the Bridge

The jumping bus is not the only daring stunt this bridge has seen. Here is one even bolder, and it happened in the sky. Picture it. The year is nineteen twelve, only eighteen years after the bridge opened, and flying machines were brand new. Most people had never seen an aeroplane in their whole lives. They were flimsy, rattly things made of wood, wire, and cloth, and getting into one at all was an act of courage.

One summer day, a pilot named Frank McClean took off in a seaplane, an aircraft with floats instead of wheels so it could land right on the water. He flew up the River Thames, low over the city, engine buzzing like an enormous angry bee. And when he reached Tower Bridge, he did something almost nobody would dare. Instead of flying safely over the top, he aimed his little plane straight at the gap between the high walkways and the road below, and flew right through the middle of the bridge.

Think about how narrow that is. Above him, the walkways. Below him, the road. On either side, the great stone towers. He had to thread his rattly wooden aeroplane through that opening like passing a needle through the eye of a thread, with barely any room to spare, and no second chances. One small mistake and he would have clipped the bridge or splashed into the river. But he made it, clean through, and then flew on under the next bridges further up the river, skimming just above the water.

People on the ground could hardly believe their eyes. A flying machine, threading Tower Bridge! It became a famous story straight away. Years later, Frank McClean said he had partly done it to make a point, to show the government how exciting and important flying could be, back when many people still thought aeroplanes were a silly toy.

So as you stand up here on the walkway, look at the open space beneath your feet, between where you are and the road far below. More than a hundred years ago, a brave man in a wooden seaplane flew a real aircraft straight through that very gap. The bus jumped the bridge, but Frank McClean flew through the middle of it.

### The Jet That Dared

More than fifty years after Frank McClean threaded his little seaplane through Tower Bridge, someone did it again, but this time in a screaming jet fighter. This story is a mix of thrilling and cheeky, so hold on tight.

The year was nineteen sixty-eight. A Royal Air Force pilot named Alan Pollock was cross. It was the fiftieth birthday of the RAF, the air force he loved, and he thought the bosses had planned nothing at all to celebrate. No flypast, no display, nothing. He felt that was disrespectful to all the pilots. So Alan Pollock decided, all by himself and without asking permission, to throw the air force a birthday party in the sky over London.

He roared low across the city in his Hawker Hunter, a fast, sleek jet, zooming past famous landmarks. And then he saw Tower Bridge ahead. On the spur of the moment, he aimed his jet at that same narrow gap between the walkways and the road, and flew straight through it, thundering between the towers at hundreds of miles an hour. Other daredevils had flown through the bridge before in slower planes, but Alan Pollock was the very first person ever to do it in a jet.

Now, flying an air force jet through Tower Bridge without permission is, as you might guess, very much against the rules. When he landed, Alan Pollock was arrested. He never flew for the RAF again. But here is the funny part. Lots of ordinary people secretly thought it was wonderful, a brave and cheeky salute to the air force on its birthday, and he became a bit of a legend. He lived a long life and only passed away in twenty twenty-five, at eighty-nine years old, still remembered as the man who flew a jet through Tower Bridge.

So this bridge has been leapt by a bus, threaded by a seaplane, and blasted through by a fighter jet. Stand here and look at that gap once more. It has tempted the bold and the reckless for over a hundred years. Please do not get any ideas of your own. Some records are best left to the daredevils of the past.

## The Bridge That Loves the Movies

Here is a job Tower Bridge does that has nothing to do with letting ships through. It is a movie star. Filmmakers absolutely love this bridge, and once you start noticing it, you will spot it again and again on screens for the rest of your life.

Think about what a director wants when a story is set in London. They want the audience to know, in a single second, exactly where they are, no words needed. And nothing says London faster than these two towers with the road that splits and lifts. Show the Eiffel Tower and everyone thinks Paris. Show Tower Bridge and everyone thinks London. So directors reach for it constantly, in spy films, cartoons, superhero adventures, and monster movies.

And here is the slightly wicked part. Because the bridge is so famous, filmmakers adore pretending to smash it up. On screen, Tower Bridge has been blown up, crashed into, frozen, flooded, and battled across by heroes and villains swinging between the towers. Giant creatures have stomped over it. Cars have gone flying off it. Of course, none of that ever really happens. It is all clever tricks, models, and computers. The real bridge is perfectly fine, opening its road for boats the very next morning as if it had never been in a single explosion. But there is something funny about a hard-working Victorian machine spending its spare time being destroyed in the movies for fun.

So why this bridge and not one of the plainer ones nearby? Partly because it is beautiful, with those fairy-tale towers, and partly because it moves. A bridge that can split open and lift into the air is a gift to any action scene. Imagine the drama of a chase when the road suddenly starts tilting upward beneath the wheels, exactly like it once did under Albert Gunter's number seventy-eight bus.

Next time you are watching a film and a city with a big river appears, keep your eyes peeled. If you spot two grey towers and a road that opens in the middle, give a little cheer. You have been there. You stood on its glass floor and looked down at the drop. You know its secrets, its steam engines, its flying bus and its daredevil pilots. To most of the audience it is just scenery. To you, it is an old friend.